

LC 207 / LC 210: Course Schedule

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Target Companies: Goldman Sachs, Flipkart, Swiggy

Problem Overview

- n courses
- $[a, b]$ means a must be studied before b
- o/p $\rightarrow$ T/F whether all courses can be studied

We can take only those courses whose prereq is 0 $\rightarrow$ hence INDEGREE, hence KAHN

Examples & Dry Run

Example 1 (Valid Schedule)

Graph: $0 \rightarrow 1, 0 \rightarrow 2, 2 \rightarrow 3$

Indegree Table:

Node	0	1	2	3
Indegree	0	1	1	1

Process:

- Now take 0 push to queue $\rightarrow$ done
- Indeg [1] --
- Indeg [2] --
- now since it's 1 : 0 and 2 : 0 $\rightarrow$ next pushes to the queue

Example 2 (Cycle Detection)

Graph: $0 \rightarrow 1, 1 \rightarrow 0$ (cycle)

- both have indegree 1, queue stays empty
- processed total $\neq 2 \rightarrow$ return false

KAHN TEMPLATE

1. **(S1)** build adjacency list
2. **(S2)** find indegree
3. **(S3)** push all indegree 0 nodes

Code Implementation (C++)

```
class Solution {
public:
    vector<int> findOrder(int n, vector<vector<int>>& prereq) {

        //basic crux ye he ki
        //agar kisika prereq nhi he to wo course le skte
        //prereq nhi matlb indeg 0
        vector<vector<int>> adj(n);
        vector<int> indeg(n,0);

        for(auto &it:prereq){
            int u=it[1];
            int v=it[0];

            adj[u].push_back(v);
            indeg[v]++;
        }

        queue<int> q;
        for(int i=0;i<n;i++){
            if(indeg[i]==0){
                q.push(i);
            }
        }

        vector<int> ans;
        while(!q.empty()){
            int u=q.front();
            q.pop();
            ans.push_back(u);
            for(int v:adj[u]){
                indeg[v]--;
                if(indeg[v]==0){
                    q.push(v);
                }
            }
        }

        return ans;
    }
};
```

```
    }  
}  
  
vector<int> ans;  
while(!q.empty()){  
    int curr=q.front();  
    q.pop();  
    ans.push_back(curr);  
  
    for(auto &it:adj[curr]){  
        indeg[it]--; //prereq course padhne ke baad future course padhne ke clos  
        if(indeg[it]==0){  
            q.push(it);  
        }  
    }  
}  
  
if(ans.size()==n) return ans;  
else return {};  
}  
};
```